

10(**short**) Figure 7 shows the geometry of an oscillating engine, where the cylinder AB rocks as the link BC rotates continuously. Suppose that BC rotates at constant $\omega = \pi \text{ rad s}^{-1}$. At $t = 0$, $\theta = 0$ and x is at its minimum value.

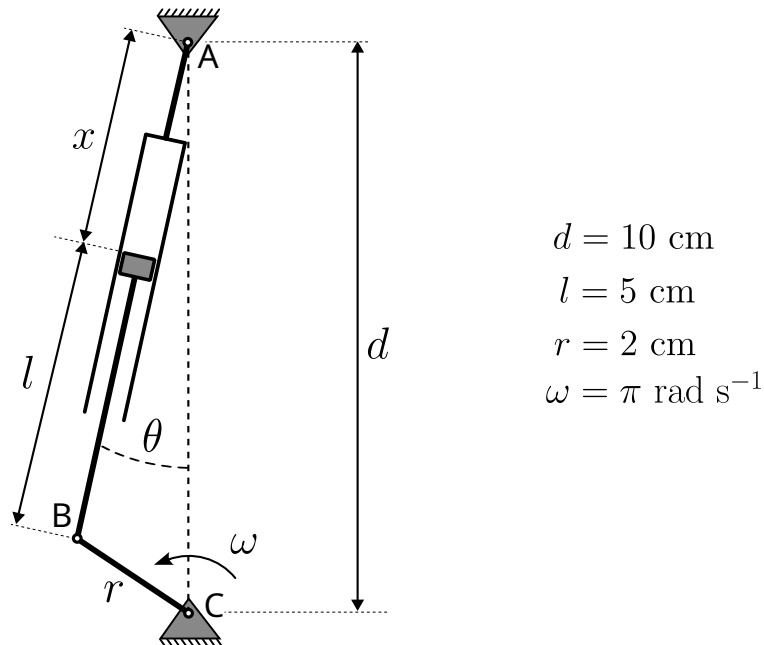


Fig. 7

- (a) Sketch the functions $x(t)$ and $\theta(t)$ for $0 \leq t \leq 2 \text{ s}$. Calculate the minimum and maximum values of x and θ and the times at which they are reached. Show these positions on your sketches. [7]
- (b) Calculate $\dot{x}$ and $\dot{\theta}$ at $t = 0$. [3]